SIMATS ENGINEERING
ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1506

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 50**
Question Count: 5

Code of Conduct

I M.Mary Sharlin (192511330) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: M.Mary Sharlin

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

ANSWERS:

1. Compare Type-1 and Type-2 Hypervisors

Type-1 and Type-2 hypervisors are used to create virtual machines, but they differ in performance, security, and management.

Type-1 Hypervisor

- Runs directly on the physical hardware.
- Provides better performance because there is no host operating system.
- Offers higher security with a smaller attack surface.
- Suitable for enterprise and cloud datacenters.
- Examples: VMware ESXi, Microsoft Hyper-V.

Type-2 Hypervisor

- Runs on top of a host operating system.
- Has lower performance due to additional software layers.
- More vulnerable because the host OS can be attacked.
- Easy to install and suitable for testing or personal use.
- Examples: Oracle VirtualBox, VMware Workstation.

Analysis
For mission-critical cloud applications, Type-1 hypervisors are the better choice because they provide higher speed, better security, improved reliability, and easier management of multiple virtual machines.

2. Virtualization Architecture and Workload Isolation

A cloud provider hosts finance, healthcare, and student portal applications on the same hardware.

Virtualization creates separate virtual machines for each workload. Each VM has its own operating system, memory, CPU allocation, and storage, preventing one application from affecting another.

Benefits of Isolation

- Prevents unauthorized access between workloads.
- Improves security and privacy.

- Reduces the impact of application failures.
- Allows independent resource allocation.

Major Attack Surfaces

1. Hypervisor Attack
   - If the hypervisor is compromised, all virtual machines may be affected.
   - Regular updates and security patches reduce this risk.
2. VM Escape Attack
   - An attacker escapes from one VM to access the host or other VMs.
   - Strong isolation and access controls help prevent this attack.

Thus, virtualization provides secure workload separation while maintaining efficient resource sharing.

3. VM Host Utilization Analysis

The VM host has the following values:

- CPU: 92%
- Memory: 88%
- Storage I/O Latency: 35 ms
- VM Boot Time: 170 seconds

Analysis

- CPU usage is very high, causing slower processing.
- Memory usage is close to maximum, leading to performance degradation.
- High storage latency delays data access.
- Long VM boot time indicates overloaded resources.

Likely Bottlenecks

- Insufficient CPU resources.
- Limited available memory.
- Slow storage subsystem.
- Too many virtual machines on one host.

Corrective Actions

- Increase CPU and RAM resources.
- Upgrade to SSD or faster storage.
- Balance workloads across multiple hosts.

- Remove unused virtual machines.
- Enable auto-scaling and continuous monitoring.

These improvements will reduce latency, improve VM startup time, and increase overall system performance.

4. Software Defined Datacenter (SDDC)

A Software Defined Datacenter manages compute, storage, and networking through software instead of physical hardware.

Compute Virtualization

- Creates multiple virtual machines on one server.
- Enables fast deployment and efficient resource utilization.

Storage Virtualization

- Combines multiple storage devices into a single storage pool.
- Simplifies backup, recovery, and storage expansion.

Network Virtualization

- Creates virtual networks without changing physical hardware.
- Allows quick network configuration and better security.

Advantages over Traditional Datacenters

- Faster deployment of resources.
- Automatic scaling based on demand.
- Lower operational cost.
- Centralized management.
- Better flexibility and disaster recovery.

Thus, SDDC provides greater agility, automation, and efficient resource management than traditional datacenters.

5. Multi-Cloud Federation and Identity Management

A multi-cloud federation project fails because different cloud providers use different identity systems.

Problem Analysis

- Users need separate login credentials for different cloud providers.
- Different authentication methods create identity mismatches.
- Poor synchronization increases security risks and management complexity.

Secure Identity Design

- Implement Identity Federation to allow users to log in once and access multiple cloud services.
- Use Single Sign-On (SSO) for a unified authentication process.
- Apply Multi-Factor Authentication (MFA) for stronger security.
- Use Role-Based Access Control (RBAC) to provide permissions based on user roles.
- Encrypt identity information to protect user privacy.
- Maintain audit logs to monitor login activities and detect suspicious access.

Benefits

- Secure authentication.
- Easier user management.
- Better privacy protection.
- Improved interoperability across multiple cloud providers.
- Reduced administrative effort and enhanced compliance.

Thus, identity federation with SSO, MFA, RBAC, and encryption ensures secure and efficient multi-cloud access.

Analytical Problem Solving Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 – Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis

		clear cause-effect analysis			
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded: